LESSON PLAN

COMPUTER (26)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	02
Total No. of Lesson Plans	10	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Conception of transmission and reception Modulation definition, need for modulation, types of modulation
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Conception of transmission and reception Modulation definition, need for modulation, types of modulation
5. Specific Objectives	26.1.1.1 Trainee will be able to understand the basic theory of communication.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of basic communication theory concepts; Safety: None; Supervision: Instructor monitors understanding of modulation necessity
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the basic function of a communication system?
(b) Why is direct transmission of low-frequency signals impractical?
(c) Can you define the term 'modulation' in your own words?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Conception of transmission and reception • Basic theory of communication systems • Signal generation and propagation basics • Role of transmitter and receiver • Importance of antenna systems	L & Ds	PPT, IB	15 min
(b)	• Definition of modulation • Need for modulation in radio transmission • Advantages of frequency shifting • Power efficiency and bandwidth considerations • Antenna size reduction requirements	L & Ds	PPT, PDF	20 min
(c)	• Amplitude Modulation (AM) basics • Frequency Modulation (FM) basics • Phase Modulation (PM) basics • Comparison of modulation types • Selection criteria for specific applications	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain why modulation is necessary for efficient radio transmission.
(b) Differentiate between AM and FM based on the carrier wave property modified.
(c) List three advantages of using modulation over direct transmission.

12. Home Assignment and Follow Up	Prepare a brief note comparing the need for modulation in LP 02 (Radar systems) versus standard communication systems.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Radar system introduction to radar system, primary and secondary radar system	Refer to the master syllabus reference list	LP 03

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min